



**AIR UNIVERSITY**  
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IBD/AU/NCCS/NCFL/369/1/08

21 October, 2025

### TO WHOM IT MAY CONCERN

1. This is to certify that Mr. Muhammad Abubakar, a student of NUML Islamabad, Pakistan, successfully completed his internship under my supervision from 7<sup>th</sup> July to 7<sup>th</sup> September 2025 on the project titled "Image Sentiment and Contextual Analysis Tool."
2. During the internship, he demonstrated strong problem-solving skills, professionalism, and timely delivery of results. Abubakar contributed to project setup, dataset collection, and FastAPI backend development, defining machine learning inference endpoints for image-based sentiment and contextual analysis. He applied image forensics techniques such as metadata extraction, Error Level Analysis (ELA), hash checks, and tampering detection (copy-move, splicing, noise, and lighting analysis). He also explored advanced methods including Photo Response Non-Uniformity (PRNU), Color Filter Array (CFA) artifacts, GAN image detection, and face morphing detection. Additionally, he supported module integration, dashboard enhancement, and presented his work clearly during the final evaluation.
3. I am confident that Mr. Muhammad Abubakar will continue to excel in his academic and professional pursuits, given his technical competence, dedication, and ability to work independently.

(DR ZUNERA JALIL)  
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